

EXAM 7 – SPRING 2011

9. (2.25 points)

Given the following triangle of cumulative paid loss data (\$000):

Accident	<u>Age of Development in Months</u>				
<u>Year</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>	<u>60</u>
2006	510	605	583	592	607
2007	543	613	550	654	
2008	552	526	535		
2009	477	630			
2010	509				

- Assume the dispersion parameter, ϕ , is 1.25 for the negative binomial model.

a. (1.25 points)

Use the negative binomial model to calculate the mean and variance of $C_{2,2}$, the incremental claim amount for accident year 2007 from 12 to 24 months.

b. (0.75 point)

Explain why it is not appropriate to fit a negative binomial model to the full triangle above.

c. (0.25 point)

Identify a model that is appropriate to fit to the full triangle which would reproduce the chain ladder unpaid claim estimate.

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Sample Solutions
Exam 07, 2011, Question 9

Solution 1

a)

$$E[C_{2,2}] = (\lambda_2 - 1)D_{2,1} = (1.140 - 1) \times 543,000 = \underline{76,156}$$

$$\text{Var}[C_{2,2}] = \phi \lambda_2 (\lambda_2 - 1) D_{2,1} = 1.25 \times 1.140 \times (1.140 - 1) \times 543,000 = \underline{108,545}$$

$$\lambda_2 = \text{Vol wgted 12-24 ldf} = (605+613+526+630)/(510+543+552+477) = 1.140$$

b)

$$\text{Vol wgted 24-36 ldf} = (583+550+535)/(605+613+526) = 0.956 < 1.000$$

The variance of the incremental claims is $\text{Var}[C_{ij}] = \phi \lambda_j (\lambda_j - 1) D_{i,j-1}$

If $\lambda_j < 1.000$, that implies a negative variance, which is not possible.

Or it's not appropriate to use negative binomial for full triangle because there are negative incremental claims existing in the triangle which implies a negative variance and is not possible.

c)

Normal approximation to the negative binomial model